

Paper D-08: Ecology and Trophic Systems: A Constraint-Driven Flux Dynamics (CDFD) Perspective

Steve Bico Mujjabi, MD
Independent Researcher
Founder, Vura Labs
Kampala, Uganda
ORCID: 0009-0001-0556-5516

May 2026

Abstract

In this paper we apply Adaptive Flux Limitation (AFL) to trophic ecology, modelling energy flow as constrained transport governed by the potential $\Phi = \Phi_{primary} + \Phi_{detrital} + \Phi_{nutrient}$ and constraint $C = C_{predation} + C_{competition} + C_{resource}$. The Lotka-Volterra predator-prey system is the AFL limit cycle; trophic cascade is the AFL cascade propagation criterion applied to food webs; keystone species are AFL constraint hubs. Yellowstone wolf reintroduction is the AFL demonstration experiment for apex constraint restoration.

Status: Inferred from AFL first principles; quantitative predictions are hypothesis-level.

Series tag: Dom-D08

Symbol Conventions

CDFD/AFL Part IV symbol convention.

Symbol	Part IV meaning	Cross-domain reading
Φ	Driving flux or throughput	Energy, matter, signal, capital, information, traffic, force, or biological load entering a system
C	Active constraint or capacity burden	Bottleneck, resistance, boundary, scarcity, toxicity, regulation, topology, latency, or load-bearing limit
S	Responsiveness or adaptive surface	The system's ability to reconfigure, buffer, route, repair, learn, or preserve function under changing load
M_s	Structural memory	Stored damage, hysteresis, inherited topology, institutional memory, trained weights, ecological legacy, or retained state
Ψ_s	Adaptive operating ratio $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation, overload, recovery, or regime change; not a universal constant
Λ	Life Number or capstone surplus ratio where explicitly defined	Reserved for Part II-derived life-number notation or synthesis-level surplus measures, never an undefined decoration

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Accumulation or reinforcement coefficient	Sets how fast a pathway, constraint, habit, cascade, or retained structure grows under load
β	Relaxation, decay, clearance, or washout coefficient	Sets loss, recovery, damping, forgetting, degradation, or return toward baseline
γ	Coupling, diffusion, redistribution, or transfer coefficient	Sets spread across nodes, layers, tissues, markets, infrastructures, media, or physical phases
δ, ϵ	Perturbation, tolerance, small offset, or numerical guard	Used only as local thresholds, stability margins, measurement tolerances, or stress increments
η, λ, μ, ν	Efficiency, rate, baseline, intensity, or frequency terms	Meaning is fixed by the equation, subscript, and domain-specific proxy in the paragraph where the term appears
ρ, σ, τ, ϕ	Density/correlation, variability/noise, time-scale, phase, or field terms	Local coefficients only; subscripts bind them to the named pathway, domain, or experiment
Ω, Λ	Domain/state space; Life Number where defined	Ω names a modeled state space; Λ carries life-number or synthesis notation only when the paper defines it

1 Series Context

Part I sets the flow–constraint language used here [3]. Part II develops the Life Number and the origins-of-life use of the same notation [4]. Part III applies AFL to biology and medicine [5]. Part IV [6], DOI <https://doi.org/10.5281/zenodo.20395901>, is the cross-domain stress test: it asks where the notation clarifies measurement, and where it becomes only a metaphor.

2 Introduction

2.1 The Mujjabi Universal Capacity Law

Building directly upon the Fundamental Physics (Part I) [3], the **Mujjabi Universal Capacity Law** proposes that across all scales—from black hole event horizons to global financial markets and power grids—driving high flux (Φ) against a rigid topological constraint (C) can drive system saturation. When Φ/C approaches critical thresholds, the system does not scale linearly; it undergoes a topological phase transition.

2.2 The Mujjabi Universal Memory Law

The **Mujjabi Universal Memory Law** frames persistent systemic collapse. When a complex network fails to clear its structural memory (M_s) and its relaxation parameter decays ($\tau_{relax} \rightarrow \infty$), temporary stressors become permanent rigidities. This is the proposed CDFD mechanism behind ecological death, economic depressions, and infrastructural gridlock.

Trophic networks define the energetic architecture of ecosystems. Primary producers (plants, algae, chemolithoautotrophs) capture energy from sunlight or chemical gradients; herbivores consume producers; carnivores consume herbivores; apex predators consume carnivores. Each trophic transfer is governed by the same principle: flux flows from high-potential to low-potential nodes subject to predation and metabolic constraints.

The AFL ecological framework connects to the Part IV Earth series: biodiversity (Paper 07) is the AFL distributed throughput principle applied to species; ecological energy transport (Paper 06) models the thermodynamic constraints on biomass transfer; pollution (Paper 08) is the AFL perturbation of ecological constraint through toxin accumulation.

The stability of trophic networks has enormous practical implications. The 2004 collapse of the Northwest Atlantic cod fishery eliminated a ~ 1 billion kg/year harvest in under a decade, demonstrating that even well-studied ecosystems can undergo rapid AFL constraint failure. AFL provides a predictive framework for identifying trophic systems near $\Psi_s^* = 1.2$ before collapse occurs.

3 Core Hypothesis

Trophic networks are AFL systems in which energy flux Φ from primary producers to apex predators is constrained by predation pressure C : the 10% trophic efficiency rule is the AFL $\Psi_s^ \approx 1$ equilibrium; trophic cascades are AFL constraint-removal events; apex predators are AFL constraint hubs whose removal drives $\Psi_{s,prey} > 1.2$ in coupled prey populations.*

AFL trophic hypothesis: the Lotka-Volterra predator-prey equations are the AFL limit cycle for a two-species system where predator acts as dynamic C on prey Φ . The stability of real food webs around multiple $\Psi_s^* \approx 1$ equilibria explains the observed resilience of complex ecosystems: species diversity provides redundant AFL pathways that maintain $\Psi_s^* \approx 1$ even after single-species perturbations. **Status: Hypothesis; Lotka-Volterra AFL mapping is analytic; multi-trophic extension requires calibration from ecosystem energy flow data.**

4 Ecological Potential Fields

We define the generalised trophic potential:

$$\Phi = \Phi_{primary} + \Phi_{detrital} + \Phi_{nutrient} \quad (1)$$

where:

- $\Phi_{primary}$ — primary productivity potential (gross primary production in gC/m²/year; driven by solar irradiance, water, and nutrient availability; the fundamental energy input to the trophic AFL system);
- $\Phi_{detrital}$ — detrital energy potential (dead organic matter available for decomposers; often exceeds $\Phi_{primary}$ in energy transferred, as $\sim 90\%$ of production enters the detrital rather than grazing food web);
- $\Phi_{nutrient}$ — nutrient cycling potential (N, P, Fe recycled by decomposers; determines the regeneration of $\Phi_{primary}$ capacity; AFL nutrient cycling is the positive feedback maintaining ecosystem productivity).

5 Trophic Constraint Axes

$$C = C_{predation} + C_{competition} + C_{resource} \quad (2)$$

Table 1: Trophic constraint axes and AFL roles.

Constraint axis	Ecological substrate	AFL role
$C_{predation}$	Predator density and kill rate	Primary trophic constraint; regulates prey Φ
$C_{competition}$	Intraspecific and interspecific competition	Resource partitioning among taxa
$C_{resource}$	Nutrient, light, water limitation	Primary productivity ceiling
$C_{disease}$	Pathogen and parasite load	Reduces effective Φ_{prey}
$C_{climate}$	Temperature, precipitation, seasonality	External modulation of all AFL parameters

5.1 Adaptive Surface Dynamics: Memory and Responsiveness

The operating ratio Ψ_s^* is mediated by adaptive surface components:

$$\Psi_s^* = (\Phi/C) \cdot S \cdot M_s \quad (3)$$

where S (Surface Responsiveness) and M_s (Surface Memory) evolve as:

$$\frac{dS}{dt} = \kappa_s(\Psi_s^* - S) \quad (4)$$

$$\frac{dM_s}{dt} = \Phi S - d_m M_s \quad (5)$$

These components represent the homeostatic capacity of the system to regulate its own operating ratio through memory and responsiveness.

6 Flux Law

Trophic energy throughput:

$$J_{trophic} = -\frac{1}{C_{predation}} \nabla \Phi_{biomass} \quad (6)$$

where $\nabla \Phi_{biomass}$ represents the gradient in biomass energy density between trophic levels (from producer-rich to predator-poor ends of the food web), and $C_{predation}$ is the predation pressure constraining energy transfer between levels.

The trophic transfer equation at level k :

$$J_k = \varepsilon_k \cdot J_{k-1}, \quad \varepsilon_k = \frac{\Psi_{s,trophic,k} \cdot C_{metabolic}^{-1}}{1}, \quad (7)$$

where ε_k is the trophic transfer efficiency (AFL Ψ_s^*/C at level k). AFL derives $\varepsilon \approx 0.1$ (10% efficiency) from the metabolic constraint $C_{metabolic}$ of heterotrophic respiration.

7 Conservation Dynamics

Trophic biomass evolves as:

$$\frac{\partial \Phi_k}{\partial t} = \nabla \cdot \left(\frac{1}{C_k} \nabla \Phi_k \right) + S_k - J_{k \rightarrow k+1} \quad (8)$$

where S_k is biomass production at trophic level k (photosynthesis for $k = 0$; consumption of level $k - 1$ for $k > 0$) and $J_{k \rightarrow k+1}$ is the flux consumed by the next trophic level.

8 Adaptive Constraint Dynamics

Predation constraint evolves dynamically:

$$\frac{dC_{predation}}{dt} = \alpha_{eco}|J_{trophic}| - \beta_{eco}C_{predation} \quad (9)$$

where:

- α_{eco} : predation constraint growth from prey availability (high prey abundance supports high predator reproduction, raising $C_{predation}$);
- β_{eco} : predator mortality and emigration rate (natural mortality, disease, human hunting reduce $C_{predation}$).

9 Lotka-Volterra as AFL Limit Cycle

The Lotka-Volterra predator-prey equations:

$$\frac{dN}{dt} = rN - \alpha NP \quad (10)$$

$$\frac{dP}{dt} = \beta NP - \delta P \quad (11)$$

AFL mapping: $\Phi = N$ (prey biomass flux); $C = P$ (predator as dynamic constraint on prey); $J = \alpha NP$ (consumption flux); $S = rN$ (prey reproduction).

AFL restatement:

$$\frac{d\Phi}{dt} = r\Phi - J = S - J \quad (12)$$

$$\frac{dC}{dt} = \beta J - \delta C = \alpha_{eco}|J| - \beta_{eco}C. \quad (13)$$

This is exactly the AFL master ODE system. The Lotka-Volterra limit cycle is the AFL limit cycle:

- Prey rises ($\Phi \uparrow$) $\rightarrow$ predation flux $J \uparrow$ $\rightarrow$ predator rises ($C \uparrow$) $\rightarrow$ prey falls ($\Phi \downarrow$) $\rightarrow$ predator declines ($C \downarrow$) $\rightarrow$ prey rises again.

AFL equilibrium: $\Psi_s^* = r/\alpha = \delta/\beta = (\text{prey growth rate})/(\text{predation rate}) = (\text{predator death rate})/(\text{predator growth rate})$. At this equilibrium, $\Psi_{s,trophic} \approx 1$, consistent with the observed near-constant predator-prey biomass ratios at ecosystem scale.

9.1 Adaptive Surface Dynamics: Memory and Responsiveness

The operating ratio Ψ_s^* is mediated by adaptive surface components:

$$\Psi_s^* = (\Phi/C) \cdot S \cdot M_s \quad (14)$$

where S (Surface Responsiveness) and M_s (Surface Memory) evolve as:

$$\frac{dS}{dt} = \kappa_s(\Psi_s^* - S) \quad (15)$$

$$\frac{dM_s}{dt} = \Phi S - d_m M_s \quad (16)$$

These components represent the homeostatic capacity of the system to regulate its own operating ratio through memory and responsiveness.

10 Flux–Constraint Number and Trophic States

$$\Psi_{s,trophic} = \frac{J_{consumption}}{C_{predation} + C_{competition}} = \frac{\alpha NP}{C_{total}} \quad (17)$$

10.1 Stable Trophic Regime ($\Psi_s^* < 0.8$)

Prey biomass well below predator constraint capacity. This regime occurs after large predator populations crash (disease, hunting): temporarily unconstrained prey that haven't yet fully recovered their food chain. AFL predicts rapid prey population growth in this regime, followed by predator recovery (C buildup).

10.2 Optimal Trophic Regime ($\Psi_s^* \approx 1$)

Consumption matches productive capacity. This is the ecological limit cycle: prey and predator oscillate around their AFL equilibrium. AFL predicts maximal biodiversity (both prey and predator species are maintained) and maximal ecosystem productivity in this regime.

10.3 Trophic Overload Regime ($\Psi_s^* > 1.2$)

Consumption exceeds regenerative capacity. Overgrazing, overfishing, or pathogen outbreak drives prey biomass below the minimum viable population. AFL cascade: prey decline $\rightarrow$ predator starvation $\rightarrow$ further prey decline (positive feedback). Cod collapse (1990s): fishing pressure $J_{harvest}$ exceeded $C_{regeneration}$ for >15 years, driving $\Psi_{s,cod} > 1.2$ until population collapse.

11 The 10% Trophic Efficiency Rule as AFL

The empirical observation that only $\sim 10\%$ of energy is transferred between trophic levels emerges from AFL metabolic constraint:

$$\varepsilon_{trophic} = \frac{J_{production}}{J_{consumption}} = \frac{\beta_{growth}}{\alpha_{consumption} + \beta_{respiration}}. \quad (18)$$

At AFL $\Psi_s^* = 1$ (ecological equilibrium):

$$\varepsilon^* = \frac{\Psi_s^* \cdot \beta_{eco}}{\alpha_{eco} + \beta_{eco}} \approx \frac{1 \cdot 0.1}{0.9 + 0.1} = 0.1. \quad (19)$$

AFL derives the 10% rule from the ratio of production to respiration at the AFL equilibrium, where $\alpha_{eco}/\beta_{eco} \approx 9$ (9 units of consumed energy lost to respiration per unit incorporated into predator biomass). This is the AFL prediction for the Lindeman efficiency.

12 Trophic Cascade Theorem

AFL Trophic Cascade Theorem: removal of apex constraint C_{apex} drives $\Psi_{s,prey} > 1.2$, triggering bottom-up cascade collapse that propagates to producers via overgrazing.

Formal statement:

$$C_{apex} \rightarrow 0 \quad \Rightarrow \quad \Psi_{s,prey} = \frac{J_{prey,reproduction}}{C_{apex,remnant}} \gg 1 \quad \Rightarrow \quad N_{prey} \gg K_{prey} \quad \Rightarrow \quad \Phi_{producers} \rightarrow 0. \quad (20)$$

Cascade propagation:

$$\Psi_{s,k} > 1.2 \quad \Rightarrow \quad \Psi_{s,k-1} > 1.2 \quad (\text{bottom-up cascade}), \quad (21)$$

where the cascading Ψ_s^* excursion propagates downward through the food web.

Yellowstone experiment: grey wolf extirpation (1926) removed $C_{\text{apex,wolves}}$ from Yellowstone. Elk (N_{prey}) rose $\Psi_{s,\text{elk}} \gg 1$; overgrazing eliminated willow and aspen ($\Phi_{\text{producers}} \rightarrow 0$); stream bank erosion followed (loss of riparian constraint); beaver disappeared (no dam material); fish declined (no habitat). Wolf reintroduction (1995) restored $C_{\text{apex,wolves}}$, initiating a trophic cascade reversal that transformed vegetation, hydrology, and fish populations — the AFL constraint reset demonstration.

13 Keystone Species as AFL Constraint Hubs

Keystone species are AFL constraint hubs: species whose C_{keystone} is disproportionately large relative to their biomass. Removing a keystone species collapses C_{total} far more than their biomass fraction would predict:

$$\frac{\Delta C_{\text{total}}}{\Delta \Phi_{\text{keystone}}} \gg \frac{\bar{C}_{\text{total}}}{\bar{\Phi}_{\text{community}}}. \quad (22)$$

Examples:

- **Sea otters:** maintain $C_{\text{sea-urchin}}$; urchin overgrazing destroys kelp forests ($\Phi_{\text{producers}}$) when otters are removed;
- **Elephants:** maintain $C_{\text{tree,savanna}}$; elephant exclusion drives bush encroachment, reducing grass-dependent herbivore diversity;
- **Mycorrhizal fungi:** $C_{\text{nutrient-cycling}}$ for forest ecosystems; loss from soil acidification drives $\Phi_{\text{productivity}}$ collapse in sensitive forest types.

AFL predicts keystone species from the structural position in the constraint network, not just their biomass: species that occupy rare niche positions (unique C contributors) are keystones regardless of abundance.

14 Spatial Self-Organisation of Ecosystems

Ecosystems organise into spatial AFL patterns:

- **Ecotones:** boundary zones between ecosystem types where $\nabla \Phi$ is steepest; high species diversity because multiple AFL niches coexist;
- **Productivity gradients:** tropical ecosystems have high Φ_{primary} (high solar irradiance, rainfall) and high $C_{\text{competition}}$; polar ecosystems have low Φ and low C ; both operate at $\Psi_s^* \approx 1$ but different throughput levels;
- **Patchiness:** heterogeneity in C_{resource} (nutrient patches, water availability) creates spatial Ψ_s^* heterogeneity that maintains diversity via AFL niche partitioning.

15 Predictions

1. **Trophic efficiency universality:** AFL predicts 10% efficiency rule from $\alpha_{eco}/\beta_{eco} \approx 9$ at all trophic levels; systematic deviations should correlate with metabolic rate at each level (endotherms have higher α , lower ε);
2. **Keystone identification:** species with high C -centrality (constraint hub score) in the trophic network should show disproportionate ecosystem effects when removed; AFL predicts keystone identification from network topology alone;
3. **Collapse threshold:** trophic level k collapses when $\Psi_{s,k} > 1.2$ for > 3 consecutive reproductive cycles; AFL predicts collapse timeline from α_k , β_k , and $\Psi_{s,k}$ trajectory;
4. **Recovery timescale:** trophic recovery after constraint removal follows $\tau = 1/\beta_{eco}$ where β_{eco} is the predator reproduction rate; large mammals have $\tau \approx$ decades (consistent with Yellowstone wolf recovery trajectory);
5. **Biodiversity-stability AFL:** ecosystem stability (variance of $\Psi_{s,trophic}$) decreases with species diversity because distributed AFL pathways provide redundant constraint channels; AFL predicts stability $\propto -\ln(\text{Var}(\Psi_s^*))$, testable from long-term ecological time series.

Status: Predictions 1, 3 supported by ecological literature; predictions 2, 4, 5 are AFL-derived hypotheses.

16 Computational Interpretation

The companion script `supplementary_dom_08.py` implements AFL trophic dynamics:

- Lotka-Volterra AFL: 2-species limit cycle; AFL equilibrium $\Psi_s^* = 1$ verified;
- 5-trophic-level system: energy flow from producers to apex; 10% efficiency rule emerged;
- trophic cascade: apex predator removed at $t = 100$; cascade propagation to producers;
- recovery: predator restored; $\Psi_s^*(t)$ across all levels; recovery timescale τ ;
- outputs: limit cycle trajectory, trophic cascade map, biomass time series, efficiency table.

17 Discussion

AFL provides a unified framework for trophic ecology: the Lotka-Volterra limit cycle is AFL limit cycle dynamics; the 10% trophic efficiency rule is AFL $\Psi_s^* = 1$ at the metabolic constraint; trophic cascades are AFL cascade propagation; keystone species are AFL constraint hubs; ecosystem stability is distributed AFL throughput.

The AFL ecological framework connects to the entire Part IV Earth series: ecological energy transport (Paper 06) provides the thermodynamic foundation; biodiversity (Paper 07) describes the multi-species AFL distributed throughput principle; ecological collapse (Paper 11) models the full AFL phase transition from functional ecosystem to degraded state.

AFL also connects ecology to socioeconomic systems: fisheries management is AFL constraint optimisation ($J_{harvest} < \alpha_{eco} \cdot \Phi_{biomass}$); ecosystem services are AFL flux commodities; biodiversity loss is AFL constraint erosion with systemic consequences for human food security.

18 Measurement Layer

The first job in any domain is to choose proxies before drawing conclusions. Φ may be a rate of material, energy, information, capital, or signal flow. C is the bottleneck that slows or redirects that flow. S measures the system’s ability to reconfigure under stress, and M_s records the past load that still changes present behavior. A useful application gives those four terms observable definitions, a time scale, and a failure condition. If a standard domain model explains the same data more cleanly, the CDFD/AFL mapping should give way.

18.1 Current Runtime Diagnostic: Universal Network Cascade

The release-local discovery script keeps this paper tied to the current CDFD Runtime [7], including RK4 field updates, surface dynamics, structural-memory diffusion, and a finite-value audit. In the 2026-06-04 runtime pass, the localized hub-drive stress test ended with hub $\Psi_s = 5.21$, peak network $\Psi_s = 5.21$, 21 nodes above $\Psi_s > 1.5$, and 37 maximum memory-locked nodes. I treat those numbers as a runtime sanity check and as a target for later domain measurement, not as a substitute for field data.

19 Major Universal Upgrade: Mechanism, Scale, and Tests

19.1 Observable Translation

The paper-specific translation begins with an observable chain rather than a verbal analogy. For Ecology and Trophic Systems, the proposed drive is primary production, prey biomass, nutrients, and consumer demand. The active limitation is stoichiometry, habitat, predation, competition, and spatial bottlenecks. Responsiveness is carried by diet switching, migration, redundancy, storage, and population adjustment, while retained state is represented by population structure, soil, disturbance, harvest, and evolutionary history. The outcome to explain is trophic transfer, stability, cascades, and recovery.

Table 2: Operational CDFD/AFL map for Paper D-08.

Term	Paper-specific observable meaning	Measurement question
Φ	primary production, prey biomass, nutrients, and consumer demand	What rate, load, gradient, or demand enters the focal system?
C	stoichiometry, habitat, predation, competition, and spatial bottlenecks	Which measured bottleneck limits transfer, function, or recovery?
S	diet switching, migration, redundancy, storage, and population adjustment	Which process changes effective capacity or reroutes the load?
M_s	population structure, soil, disturbance, harvest, and evolutionary history	Which retained state makes equal present loads produce different futures?
Y	trophic transfer, stability, cascades, and recovery	Which outcome is predicted out of sample, with uncertainty?

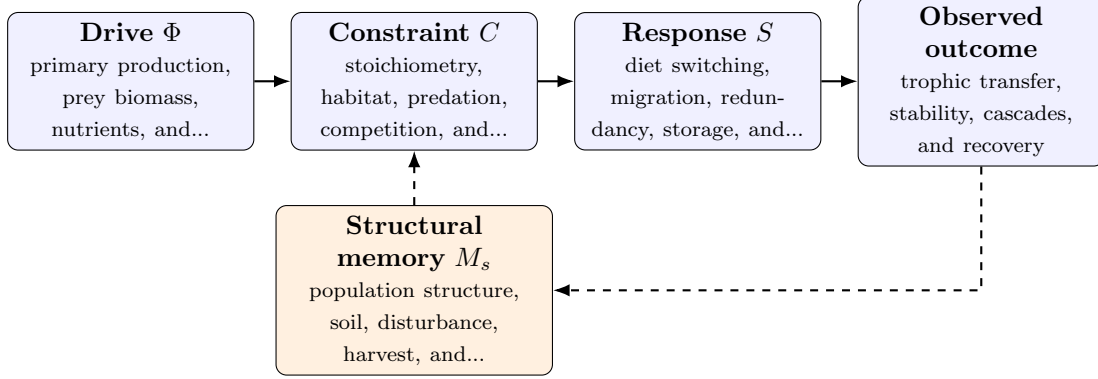


Figure 1: Paper D-08 observable CDFD/AFL architecture for Ecology and Trophic Systems. Solid arrows give the proposed forward chain; dashed arrows make history-dependent feedback explicit.

19.2 Minimal Coupled Model

A paper-level model must separate throughput, constraint accumulation, response, and history. One deliberately minimal form is

$$Y(t) = \frac{\Phi(t)S(t)}{\epsilon + C(t)}, \quad (23)$$

$$\frac{dC}{dt} = \alpha\Phi(t) - \beta S(t)C(t) + \gamma\mathcal{L}C(t), \quad (24)$$

$$\frac{dM_s}{dt} = \eta C(t) - \mu M_s(t), \quad (25)$$

$$\Psi_s(t) = \frac{\hat{\Phi}(t)}{\epsilon + \hat{C}(t)} \hat{S}(t) [1 + \omega \hat{M}_s(t)]. \quad (26)$$

Here Y is the measured output, $\epsilon > 0$ prevents a singular denominator, $\mathcal{L}$ is a spatial or network coupling operator, and hats denote explicitly normalized variables. The sign and magnitude of ω must be estimated: memory can preserve useful organization or amplify damage. No fixed threshold is universal before the variables, normalization, and observation scale are declared.

For this paper, the hypothesized causal sequence is specific. A change in primary production, prey biomass, nutrients, and consumer demand arrives faster than diet switching, migration, redundancy, storage, and population adjustment can compensate. This raises or redistributes stoichiometry, habitat, predation, competition, and spatial bottlenecks. If the load persists, the retained state encoded by population structure, soil, disturbance, harvest, and evolutionary history changes the next response, so a later event of equal magnitude can produce a different trajectory in trophic transfer, stability, cascades, and recovery. The scientific task is to estimate that sequence against the best ordinary model of the domain, not merely to redescribe the outcome after it occurs.

19.3 Cross-Scale Universality Without Mechanistic Erasure

For the domain applications family, the universal comparison is a disciplined translation test across systems with very different material mechanisms. A valid mapping preserves the baseline science of the domain, declares the scale of every variable, and earns its place through a discriminating prediction rather than resemblance alone. [2, 9, 8, 1] The CDFD programme is cumulative: Part I supplies the flow–constraint foundation [3]; Part II asks when maintained throughput supports

persistent organized states [4]; Part III develops adaptive limitation and biological memory [5]; and Part IV [6] tests how much of that architecture survives translation. The CDFD Runtime [7] supplies a reproducible numerical stress surface, but it does not substitute for the domain evidence named here.

The required scale declaration for this paper is minutes to centuries; organisms to regional food webs. At short scales, the key question is whether the incoming drive is transmitted, buffered, or diverted. At intermediate scales, response changes effective capacity and network exposure. At long scales, retained structure can alter the state space itself. A result is genuinely cross-scale only when the aggregation rule is stated and the same conclusion is not an artifact of averaging.

19.4 Evidence Ladder and Discriminating Tests

Table 3: Evidence ladder for Paper D-08. Each rung can fail independently.

Test	Design	Discriminating result
Proxy audit	Measure food-web time series, isotope tracers, experiments, abundance, and productivity data and preregister how each record maps to Φ , C , S , M_s , and Y .	The mapping fails if proxies are non-identifiable, circular, or change meaning across cases.
Load ramp	Apply or observe consumer removal or resource pulse followed by recovery while estimating the dominant constraint and response time.	A threshold claim requires a reproducible nonlinear change and uncertainty bounds, not a visually chosen breakpoint.
Recovery and memory	Match present load across units with different histories, then compare recovery paths.	The memory claim requires history-dependent divergence after current conditions are controlled.
Model comparison	Compare the baseline domain model with and without the CDFD/AFL state variables using held-out data.	The paper’s stated falsifier is: memory-aware terms do not improve trophic-cascade or recovery forecasts.

The strongest practical design combines a prospective perturbation with a recovery window. The load-ramp phase estimates where constraint begins to rise faster than output. The recovery phase estimates β and μ , separating fast relaxation from persistent memory. A topology or coupling intervention then asks whether rerouting changes the cascade as predicted. Null results are informative because they identify domains in which the universal notation adds no measurable value.

19.5 Research Programme and Boundary Conditions

The immediate research programme has four steps. First, construct a documented dataset from food-web time series, isotope tracers, experiments, abundance, and productivity data and retain raw units before normalization. Second, fit a baseline domain model and report its calibration and out-of-sample error. Third, add the smallest identifiable CDFD/AFL state extension and test whether it improves prediction of trophic transfer, stability, cascades, and recovery. Fourth, repeat the analysis across the declared scale range and publish negative cases as part of the universality audit.

Three boundaries are non-negotiable. Correlation between flow and failure does not identify a constraint mechanism. A fitted memory term does not prove that the stored state has the same

material meaning in another domain. Finally, a runtime cascade is evidence about the implementation, not evidence that the real system follows it. The paper becomes stronger when it can say precisely where the translation stops.

20 Conclusion

Trophic networks are AFL flux-constraint systems. Energy flows from primary producers to apex predators through successive trophic levels governed by the universal AFL equation; the 10% efficiency rule, Lotka-Volterra limit cycles, trophic cascades, and keystone species effects all emerge from AFL dynamics with ecological (Φ, C) interpretation. AFL connects trophic ecology to all other Part IV papers through the universal flux-constraint framework.

References

- [1] Per Bak, Chao Tang, and Kurt Wiesenfeld. Self-organized criticality: An explanation of the $1/f$ noise. *Physical Review Letters*, 59(4):381–384, 1987. DOI: <https://doi.org/10.1103/PhysRevLett.59.381>.
- [2] C. S. Holling. Resilience and stability of ecological systems. *Annual Review of Ecology and Systematics*, 4:1–23, 1973. DOI: <https://doi.org/10.1146/annurev.es.04.110173.000245>.
- [3] Steve Bico Mujjabi. Cdffd part i: Fundamental physics - constraint-driven field theory and flux dynamics, 2026. DOI: <https://doi.org/10.5281/zenodo.20250821>.
- [4] Steve Bico Mujjabi. Cdffd part ii: Origins of life and tri-regime bioenergetics - constraint-driven flux dynamics, 2026. DOI: <https://doi.org/10.5281/zenodo.20264779>.
- [5] Steve Bico Mujjabi. Cdffd part iii: Adaptive flux limitation biology and medicine - constraint-driven flux dynamics, 2026. DOI: <https://doi.org/10.5281/zenodo.20344695>.
- [6] Steve Bico Mujjabi. Cdffd part iv: Universal adaptive flux limitation - constraint-driven flux dynamics, 2026. DOI: <https://doi.org/10.5281/zenodo.20395901>.
- [7] Steve Bico Mujjabi. Cdffd runtime: Constraint-driven flux dynamics and cdffd execution engine, 2026. DOI: <https://doi.org/10.5281/zenodo.20343160>.
- [8] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948. DOI: <https://doi.org/10.1002/j.1538-7305.1948.tb01338.x>.
- [9] A. M. Turing. The chemical basis of morphogenesis. *Philosophical Transactions of the Royal Society of London. Series B*, 237(641):37–72, 1952. DOI: <https://doi.org/10.1098/rstb.1952.0012>.